

Task 2: Data Visualization and Storytelling

Superstore-style sales analysis built for Tableau or Power BI storytelling

Objective

This report turns sales data into a business story: where revenue comes from, which categories generate profit, and how discounting affects margin. The design favors clarity, context, and decision-making over decoration.

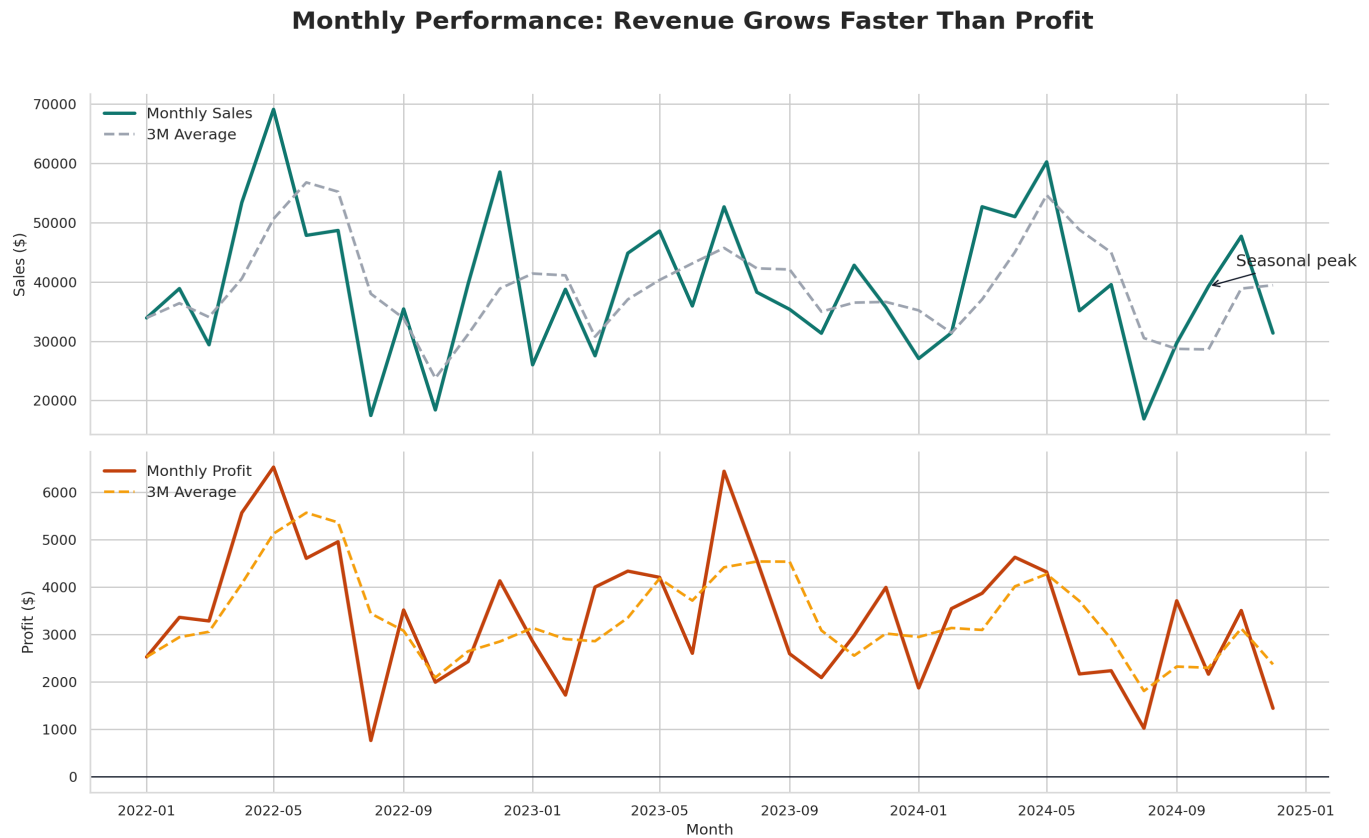
Metric	Value	Metric	Value
Orders	1,200	Revenue	\$1,410,712
Profit	\$120,497	Profit Margin	8.5%
Avg Discount	16.7%	Loss-Making Orders	25.2%

Storyline

- Profit peaked in May 2022 at \$6,528, while Aug 2022 was the weakest month at \$764.
- Technology delivered the strongest profit contribution, while Furniture lagged behind.
- East led revenue, which suggests the best opportunity is to protect its margin instead of chasing extra discounting.
- Orders with discounts of 20% or more averaged \$8 profit, compared with \$174 for low-discount orders.

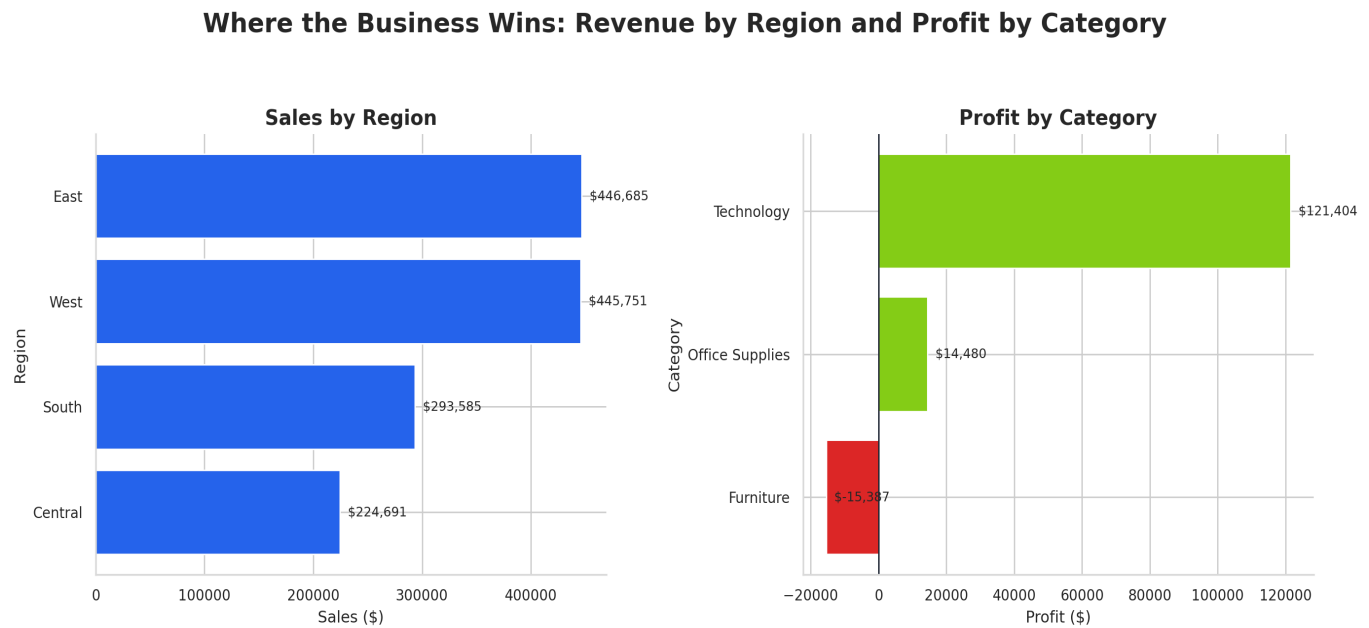
Chart 1: Monthly Revenue and Profit

Use this chart to show seasonality, momentum, and whether profit is keeping pace with sales.



Takeaway: sales rise over time, but profit is more sensitive to discounting and product mix.

Chart 2: Region and Category Performance



Takeaway: revenue is concentrated in the top regions, while category-level profit reveals where margin is strongest.

Chart 3: Discount Pressure

Discount Pressure: Higher Discounts Drag Profit Down

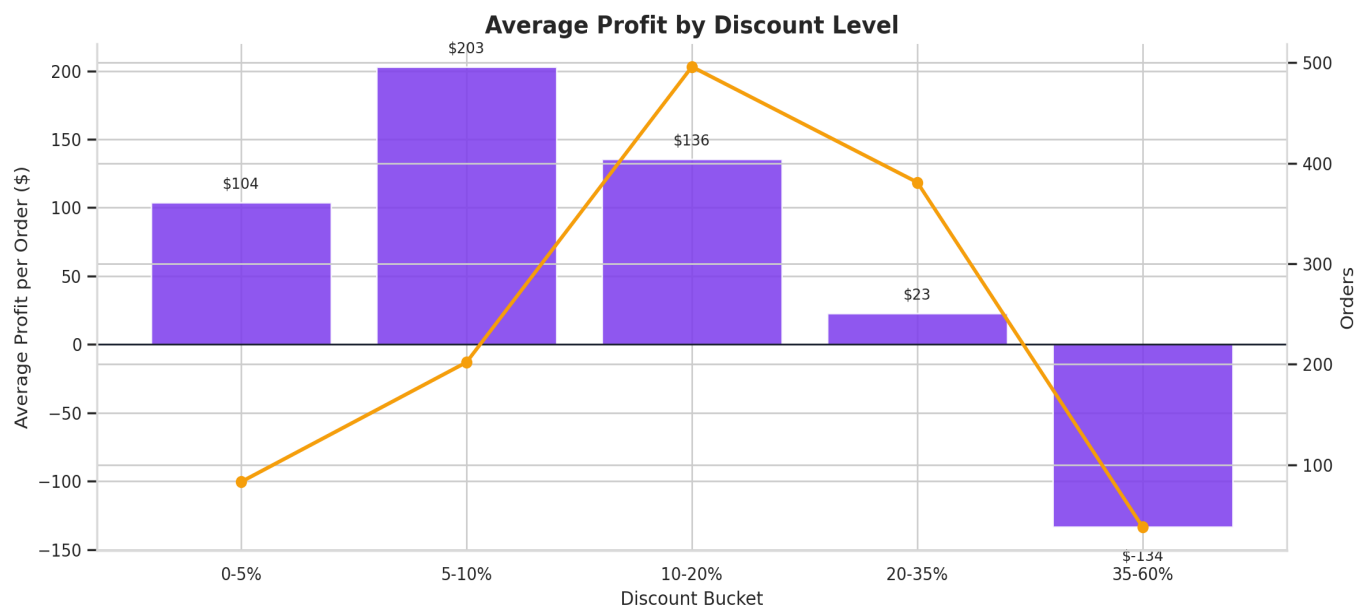
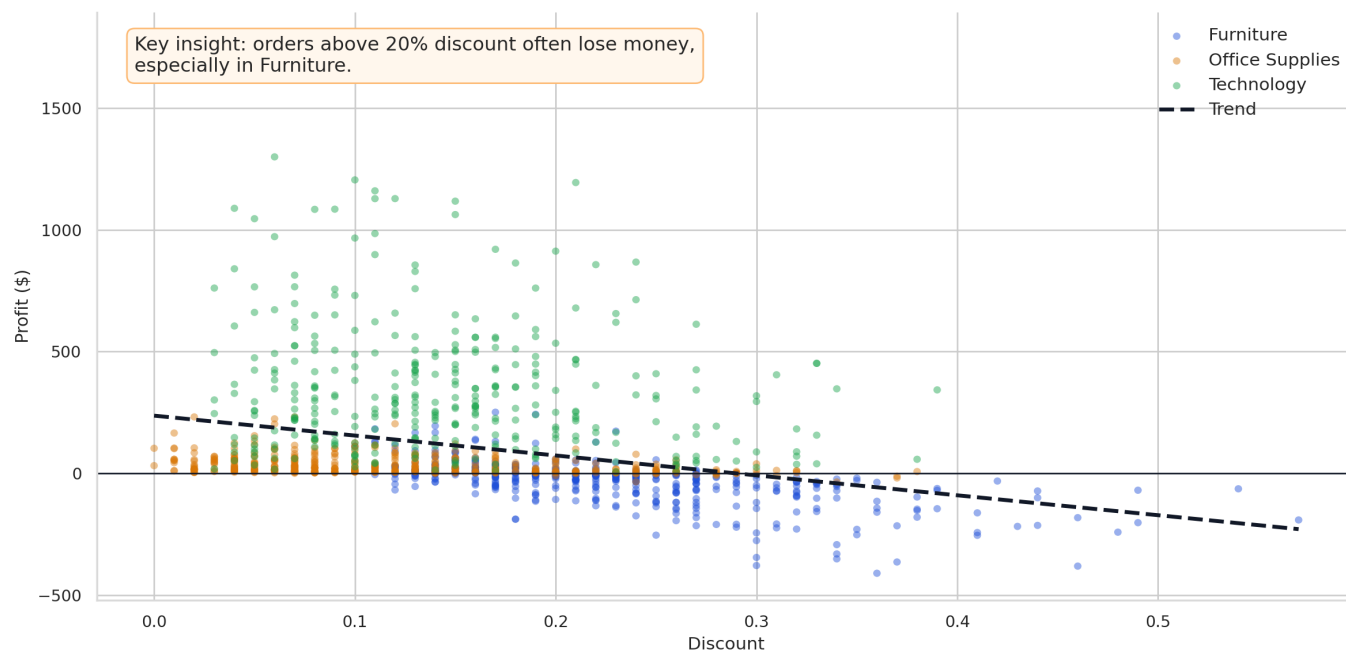


Chart 4: Discount vs Profit

Discount vs Profit: Deep Discounts Create Negative Outliers



Takeaway: once discounts move beyond the 20% range, profit erosion becomes clear.

Dashboard Storyboard

Slide	Purpose	What the audience learns
1	Executive summary	What the business did well and where it struggled
2	Sales trend	Whether growth is stable or seasonal
3	Regional view	Which markets drive revenue
4	Category view	Which products create or destroy profit
5	Discount analysis	How pricing policy affects margin
6	Action slide	What to do next

Interview Questions and Answers

What is the importance of data visualization?

It turns raw numbers into patterns that people can understand quickly, which improves decision-making and helps teams spot trends, risks, and opportunities.

When do you use a pie chart vs bar chart?

Use a pie chart only when you want to show part-to-whole for a few categories. Use a bar chart when comparing values across categories because it is easier to read and more accurate.

How do you make visualizations more engaging?

Keep the layout clean, choose one clear message per chart, use color sparingly to highlight key points, and add annotations that explain why the chart matters.

What is data storytelling?

Data storytelling combines analysis, visuals, and narrative so the audience understands not only what happened but also why it matters and what should happen next.

How do you avoid misleading visualizations?

Use correct scales, avoid unnecessary 3D effects, label axes clearly, do not truncate context, and choose the chart type that matches the data and the question.

What are best practices in dashboard design?

Focus on a single business goal, keep the most important KPIs at the top, reduce clutter, use consistent colors, and make interactions simple and purposeful.

What tools have you used for visualization?

Common tools include Tableau, Power BI, Excel, and Python libraries such as Matplotlib, Seaborn, and Plotly.